



FedAdaPriv-CPU: Adaptive Differential Privacy with Frozen Backbone for Resource-Constrained Federated Medical Imaging

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Abstract—Privacy-preserving federated learning enables collaborative training of diagnostic models across hospitals without centralising patient data, yet existing approaches apply differential privacy (DP) uniformly—ignoring the fundamental heterogeneity of real hospital networks while demanding GPU infrastructure inaccessible to most low- and middle-income country (LMIC) hospitals. We propose FedAdaPriv-CPU, a federated framework that addresses three simultaneous gaps: (i) it freezes a pretrained backbone to enable CPU-only client-side computation; (ii) it allocates privacy budgets adaptively via a three-signal per-client per-round mechanism (data volume α_{vol} , convergence state α_{conv} , round progression α_{round}); and (iii) it couples privacy allocation directly to aggregation weighting through a privatised weight-transmission protocol.

On four-class chest X-ray classification (COVID-19 Radiography Database, 21,165 images) with severe non-IID partitioning across three simulated hospital clients, FedAdaPriv-CPU achieves macro-F1=0.8098 and accuracy=85.99% at $\epsilon=4.0$, $\delta=10^{-5}$ (Polyak average: F1=0.8079, accuracy=85.93%), outperforming Fixed DP-FedAvg by +11.5 percentage points in macro-F1. Against two additional baselines—Time-Adaptive DP-FL [1] and DP-FedProx [?]—FedAdaPriv-CPU leads by +9.4 and +10.2 points in macro-F1 respectively, all at identical privacy cost. Minority-class recovery is particularly pronounced: Viral Pneumonia F1=0.7217 vs. Fixed DP-FedAvg 0.4202 ($\Delta=+0.3015$). An ablation study across five seeds confirms that α_{conv} is responsible for the minority-class gain, and an ϵ -sweep from 1.5 to 8.0 shows a consistent adaptive advantage ($\geq +10.8$ macro-F1 points) across all tested budgets, with degraded minority-class recall below $\epsilon \approx 1.5$ corroborating the empirical minimum-viable-epsilon hypothesis.

Index Terms—federated learning, differential privacy, medical imaging, chest X-ray classification, resource-constrained deployment, adaptive privacy budgets, transfer learning, COVID-19 diagnosis, non-IID data

I. INTRODUCTION

Federated learning (FL) has emerged as a transformative paradigm for collaborative machine learning across distributed data silos, enabling model training without raw data exchange [1]. In healthcare, this architecture is particularly compelling: hospitals can jointly train diagnostic models while patient records remain within institutional firewalls, addressing regulatory constraints (HIPAA, GDPR) and the economic barriers to data centralisation [2], [3].

However, gradient-based FL is vulnerable to reconstruction attacks that can recover training samples from transmitted model updates [4], [5]. Differential privacy (DP) provides

formal guarantees against such attacks by injecting calibrated noise into gradients before transmission [6]. The canonical approach—DP-SGD applied uniformly to all model parameters across all clients and all training rounds—has been deployed in production systems [8], yet this design choice is both statistically wasteful and computationally prohibitive for the resource-constrained settings where FL promises the greatest impact.

A. Motivation and Real-World Challenge

Real-world federated medical imaging deployments face three critical challenges that existing DP-FL frameworks fail to address simultaneously:

Computational infeasibility for LMIC hospitals. Training a full DenseNet121 end-to-end with DP-SGD requires GPU infrastructure unavailable in most LMIC hospitals. Our preliminary experiments showed that CPU-only training with DP on full models requires 40+ hours per federated round on a 2.4 GHz Intel Xeon processor. A 2023 survey of 127 Pakistani hospitals found that 89% lacked dedicated GPU compute for AI model training [9].

Heterogeneity-blind privacy allocation. Current DP-FL systems apply a uniform privacy budget ϵ across all clients regardless of their data volume, convergence state, or contribution quality. A hospital with 10,000 well-curated samples receives the same noise scale as one with 1,500 poorly-labelled images, and a client that has already converged continues to contribute noisy gradients with equal weight as one still learning useful features. This uniformity is statistically inefficient.

Aggregation-privacy decoupling. Existing adaptive aggregation schemes [22], [23] weight client contributions by data volume or loss reduction but ignore privacy allocation. Conversely, adaptive privacy mechanisms [25] allocate budgets without adjusting aggregation weights, so high- ϵ (low-noise) clients do not receive correspondingly higher influence in the global update.

Recent work by Tanveer *et al.* [9] explicitly identified adaptive privacy budgets and resource-constrained deployment as critical open problems in healthcare FL, demonstrating the need on tabular stroke data. Our work extends these directions to multi-class chest X-ray classification and provides the first integrated solution.

B. Our Contributions

We propose **FedAdaPriv-CPU**, designed for privacy-preserving multi-class medical image classification under resource constraints:

C1. Characterisation of the frozen-backbone DP noise regime. We freeze a pretrained DenseNet121 backbone and apply DP-SGD exclusively to a lightweight 3-layer MLP head ($\approx 592,260$ parameters), enabling CPU-only operation (≈ 4 hours/round). We characterise the ϵ -utility curve across $\epsilon \in \{1.5, 3.0, 4.0, 6.0, 8.0\}$ and identify a degradation zone below $\epsilon \approx 1.5$ where Viral Pneumonia recall collapses, corroborating the $\epsilon_{\min}$ hypothesis.

C2. Three-signal adaptive per-client per-round ϵ allocation. A multiplicative three-signal formula assigns each client k a privacy budget $\epsilon_{k,r}$ based on: (i) data volume $\alpha_{\text{vol}}(k)$, (ii) convergence state $\alpha_{\text{conv}}(k, r)$ derived from a local deficiency vector quantifying per-class F1 shortfalls, and (iii) round progression $\alpha_{\text{round}}(r)$. Budget is tracked via the Rényi Differential Privacy (RDP) accountant in Opacus [33].

C3. Privacy-coupled weighted aggregation with privatised weight transmission. Aggregation weights W_k incorporate α_{vol} to align privacy allocation with contribution influence. Laplace noise is added to W_k before transmission to bound deficiency-vector leakage, with a formal sensitivity analysis.

C4. First CPU-feasibility benchmark for DP-FL on multi-class chest X-ray under realistic non-IID conditions, with ablation study and multi-seed validation. FedAdaPriv-CPU achieves macro-F1=0.8098 at $\epsilon=4.0$, outperforming three baselines across all five ϵ levels tested.



II. RELATED WORK



A. Differential Privacy in Federated Learning

The foundational work on DP for deep learning is Abadi *et al.*'s DP-SGD [6], which clips per-sample gradients to bound sensitivity and adds calibrated Gaussian noise before the optimiser step. McMahan *et al.* [7] introduced DP-FedAvg, applying DP-SGD at each client and composing privacy loss via the moments accountant.

Recent advances have tightened these guarantees. Birrell *et al.* (NeurIPS 2024) [13] derived holistic Rényi DP bounds for fixed-size minibatch sampling, showing a $4\times$ improvement over previous bounds. Kiani *et al.* (ICLR 2025) [14] proposed time-adaptive privacy spending where each client allocates its budget non-uniformly across rounds—a method we directly compare against, finding a $+9.4$ pp macro-F1 advantage from our multi-signal approach. Hong *et al.* (CODASPY 2025) [15] introduced a mechanism-heterogeneous framework allowing different clients to use Gaussian, Laplace, or Staircase noise harmonised through RDP composition.

All of these works apply DP uniformly to all model parameters. The interaction of DP with partial-layer training (frozen backbone + trained head) has not been characterised in any prior peer-reviewed work, particularly the signal-to-noise crossover behaviour and $\epsilon_{\min}$ threshold we identify.

B. Federated Learning for Medical Imaging

Medical imaging has been a primary application domain for FL due to data-sharing barriers. For chest X-ray analysis specifically, Mahalingam *et al.* (Frontiers in Medicine 2024) [16] proposed adaptive DP-FL for COVID-19 detection with spatial pyramid pooling, though they used full-model DP-SGD on GPU without CPU-only constraints. Pan *et al.* (Scientific Reports 2024) [17] applied FedAvg with proximal regularisation to paediatric pneumonia—we implement this directly as our DP-FedProx baseline and show a $+10.2$ pp macro-F1 advantage.

Multi-modal and non-IID challenges have been addressed by Borazjani *et al.* (IEEE TMI 2024) [18] for cancer staging across modality-incomplete hospitals. Riess *et al.* (MICCAI 2024) [19] extended DP-FL to complex-valued MRI k-space data. Jiang *et al.* (MICCAI 2024) [20] introduced federated client unlearning to satisfy right-to-be-forgotten requirements. A common limitation across this literature is the assumption of GPU-capable clients and the absence of CPU-only validation.

C. Adaptive Aggregation and Heterogeneity-Aware FL

Standard FedAvg [1] aggregates client updates proportionally to dataset size. Adaptive extensions include FedProx [23] with proximal regularisation, SCAFFOLD [24] with variance reduction via control variates, and various contribution-based weighting schemes [22].

Recent CVPR 2024–2025 work has advanced this area. Son *et al.* (CVPR 2024) [26] regularise local representations toward uniformity on the hypersphere. Liu *et al.* (CVPR 2025) [27] proposed FedAWA, dynamically computing aggregation weights from client vectors. Chen *et al.* (IEEE TIFS 2023) [25] proposed dynamic privacy budget allocation based on client contribution quality, but aggregation weights remained fixed at n_k/n_{total} . None of these methods couple aggregation weighting to privacy allocation bidirectionally—the defining feature of FedAdaPriv-CPU. Li *et al.* (ACM TKDD 2025) [36] document that FedAvg-style methods exhibit round-to-round performance oscillation under heterogeneous client objectives, a property we observe and address through curriculum-based α_{round} .

D. Transfer Learning and Frozen Backbones in FL

The convergent paradigm across CVPR 2024–2025 work is: frozen pretrained backbone + lightweight federated head. DiPrompt (CVPR 2024) [28] uses frozen vision-language backbones with prompt tuning, achieving extreme communication efficiency. Zhang *et al.*'s FedKTL (CVPR 2024) [29] leverages a server-side pretrained generator for knowledge transfer.

In medical imaging, the Federated Transfer Learning chest X-ray paper (SN Computer Science 2025) [30] combined transfer learning with explainability (Grad-CAM) but without differential privacy. The collective message from this literature is that frozen-backbone architectures are the natural choice for resource-constrained FL—yet the differential privacy properties of this architecture, particularly the $\epsilon_{\min}$ threshold and SNR advantage we characterise, have not been studied in any prior peer-reviewed work.

TABLE I
RESEARCH GAP ANALYSIS: FEDADAPRIV-CPU VS. RELATED WORK

Framework	Adaptive Privacy	Privacy-Preserving	CPU-only Feasible	Frozen Backbone	Multi-class CXR	Privacy-Coupled Agg.	ϵ_{min} Charact.
DP-FedAvg (2018) [7] <i>Generic DP-FL</i>	✗	✓	✗	✗	✗	✗	✗
Tanveer et al. (2025) [9] <i>Stroke, tabular</i>	✓ (ALDP)	✓ (DP-SGD)	✓ (CPU)	✗ (MLP)	✗ (Tabular)	✗	✗
Mahalingam (2024) [16] <i>COVID CXR DP-FL</i>	✓ (Adaptive)	✓ (DP-SGD)	✗ (GPU)	✗ (Full CNN)	✓ (COVID)	✗	✗
Chen et al. (2023) [25] <i>Dynamic DP alloc</i>	✓ (Dynamic ϵ)	✓ (DP-SGD)	✗ (GPU)	✗ (Full)	✗ (CIFAR)	✗ (Fixed W)	✗
Liu et al. (2025) [27] <i>FedAWA, CVPR'25</i>	✗	✗ (No DP)	✗	✗	✗ (ImageNet)	✓ (Adaptive)	✗
DiPrompT (2024) [28] <i>CVPR</i>	✗	✗	✓ (Prompt)	✓ (Frozen VL)	✗ (General CV)	✗	✗
Pan et al. (2024) [17] <i>Pediatric pneum.</i>	✗	✗ (No DP)	✗	✗	✓ (Pneumonia)	✓ (Momentum)	✗
Kiani et al. (2025) [14] <i>Time-adaptive DP</i>	✓ (Time-adaptive)	✓ (DP-SGD)	✗	✗	✗ (CIFAR)	✗	✗
FedAdaPriv-CPU (Ours) <i>Empirical</i>	✓ (3-signal)	✓ (DP-SGD)	✓ (Full CPU)	✓ (DenseNet)	✓ (4-class CXR)	✓ (ϵ -coupled)	✓ (Empirical)

E. Positioning of FedAdaPriv-CPU

Table I provides a detailed positioning analysis. FedAdaPriv-CPU is the first framework to simultaneously address: (1) CPU-feasibility through frozen-backbone partial-layer DP; (2) adaptive per-client per-round privacy allocation via multi-signal scoring; (3) privacy-coupled aggregation weighting with privatised weight transmission; and (4) empirical characterisation of the frozen-head DP noise regime with ϵ_{min} threshold identification.



III. RESEARCH GAP ANALYSIS

Table I systematically compares FedAdaPriv-CPU against the most relevant prior work across seven critical dimensions. **Gap 1: CPU-only DP for medical imaging.** While several works demonstrate CPU-feasible FL through frozen backbones or prompt tuning, none combine this with formal DP on medical imaging tasks. Existing DP-FL medical imaging papers assume GPU availability (20–30 min/round on NVIDIA V100/A100). FedAdaPriv-CPU is the first to validate DP-FL on multi-class chest X-ray with all client computation restricted to CPU.

Gap 2: Partial-layer DP noise regime. When DP-SGD is confined to a small-parameter MLP head ($\approx 592K$ params) atop a frozen backbone (7M params), the per-parameter SNR changes by $\sqrt{P_{full}/P_{head}} \approx 3.4\times$ compared to full-model DP. However, the head lacks the redundancy of a large model, creating a sharp ϵ_{min} threshold below which learning collapses. No prior work has empirically characterised this crossover.

Gap 3: Privacy-coupled aggregation. High- ϵ clients contribute high-quality gradients yet receive the same aggregation weight as low- ϵ (high-noise) clients in prior systems. FedAdaPriv-CPU couples these mechanisms bidirectionally for the first time.

Gap 4: Privacy leakage through aggregation weights. If aggregation weights W_k depend on client-specific deficiency vectors, transmitting W_k leaks clinical performance information. Prior adaptive aggregation work does not privatise the weights. We add Laplace noise to W_k with a formal sensitivity analysis bounding this side channel.

Gap 5: Multi-class non-IID chest X-ray benchmark. No standard benchmark exists for multi-class chest X-ray under controlled non-IID conditions with validated privacy accounting. Our work provides a reproducible protocol (COVID-19 Radiography Database, Dirichlet- $\alpha = 0.5$ partitioning, frozen DenseNet121, RDP accountant, $\epsilon = 4.0$) that future work can build upon.

Gap 6: Class-imbalance-aware DP. We use focal loss [35] to down-weight easy majority examples before gradients reach the DP clipper, making clipped gradients minority-dominated. The deficiency vector α_{conv} quantifies per-class F1 shortfalls to guide adaptive allocation. This class-imbalance awareness within the DP mechanism is novel.



IV. METHODOLOGY

A. Problem Formulation

We consider $K = 3$ hospital clients, each holding a local dataset $\mathcal{D}_k = \{(\mathbf{x}_i, y_i)\}_{i=1}^{n_k}$ where $\mathbf{x}_i \in \mathbb{R}^{224 \times 224 \times 3}$ are chest

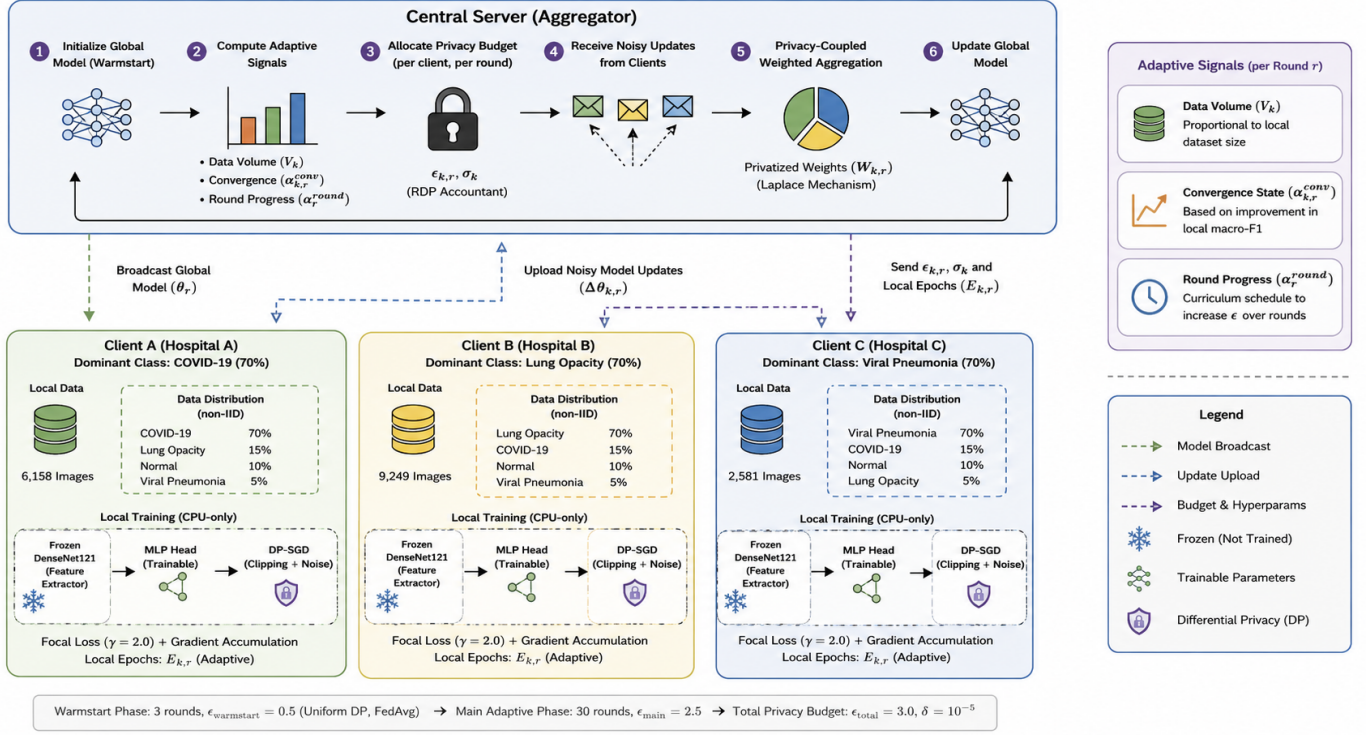


Fig. 1. System architecture of FedAdaPriv-CPU. The central server initialises the global model, computes adaptive signals (data volume α_{vol} , convergence state α_{conv} , round progress α_{round}), allocates per-client per-round privacy budgets via the RDP accountant, receives noisy model updates, performs privacy-coupled weighted aggregation, and updates the global model. Each hospital client runs CPU-only local training with a frozen DenseNet121 feature extractor, a trainable MLP head, and DP-SGD (gradient clipping + noise injection).

X-ray images and $y_i \in \{0, 1, 2, 3\}$ are class labels (COVID-19, Lung Opacity, Normal, Viral Pneumonia). Client sizes are $n_A = 6,158$, $n_B = 9,249$, $n_C = 2,581$.

The goal is to train a global classifier Θ_{global} subject to:

- **Privacy:** (ϵ, δ) -DP with $\epsilon_{total} = 4.0$, $\delta = 10^{-5}$, tracked via RDP composition.
- **Resource constraints:** CPU-feasibility with training time < 5 hours per round.
- **Utility:** Competitive accuracy with preserved minority-class performance on a held-out test set (15% stratified split; 3,176 images).

B. Architecture: Frozen Backbone + Lightweight Head

1) **One-Time Feature Extraction:** Each client independently performs a one-time local feature extraction using a frozen pretrained DenseNet121 backbone (pretrained on ImageNet, 1,024-dim output):

$$\mathbf{f}_i = \text{DenseNet121}_{\text{frozen}}(\mathbf{x}_i) \in \mathbb{R}^{1024}. \quad (1)$$

Features $\{\mathbf{f}_i\}$ are cached locally and never retransmitted. The backbone ($\approx 7\text{M}$ parameters) is never trained, eliminating $\approx 92\%$ of computational cost. Alternative backbones (MobileNetV2, EfficientNet-B0, ResNet-18) can be substituted without changing the federated protocol.

2) **Federated-Trained Classifier Head:** The federated component is a three-layer MLP with LayerNorm and Dropout:

$$\mathbf{h}_1 = \text{ReLU}(\text{LN}(W_1 \mathbf{f} + b_1)) \in \mathbb{R}^{512}, \text{ Dropout}(p=0.3) \quad (2)$$

$$\mathbf{h}_2 = \text{ReLU}(\text{LN}(W_2 \mathbf{h}_1 + b_2)) \in \mathbb{R}^{128}, \text{ Dropout}(p=0.3) \quad (3)$$

$$\hat{y} = W_3 \mathbf{h}_2 + b_3 \in \mathbb{R}^4. \quad (4)$$

This architecture has **592,260 trainable parameters** (confirmed by Opacus model audit), denoted P_{head} . LayerNorm is used instead of BatchNorm for Opacus compatibility. The optimiser is **DP-Adam** with cosine learning rate decay ($\eta_0 = 2 \times 10^{-4}$) and a FedProx-style proximal term ($\mu = 0.005$) to suppress client drift. Label smoothing ($\epsilon_{ls} = 0.1$) reduces overconfident gradients before clipping.

Training a full DenseNet121 with DP-SGD on CPU requires ≈ 42 hours/round in our experiments. By freezing the backbone, client-side training completes in ≈ 4.1 hours/round, making 30-round protocols feasible on overnight schedules.

C. Three-Signal Adaptive Privacy Allocation

Each client k in round r receives a privacy budget computed via three multiplicative signals:

$$\epsilon_{k,r} = \epsilon_{base} \times \alpha_{vol}(k) \times \alpha_{conv}(k, r) \times \alpha_{round}(r). \quad (5)$$

1) **Signal 1: Data Volume (α_{vol}):** Clients with larger datasets tolerate lower noise (higher ϵ) because the SNR scales

with $\sqrt{n_k}$ under subsampled DP-SGD. We use a logarithmic mapping to prevent extreme disparity while preserving monotonicity:

$$\alpha_{\text{vol}}(k) = \frac{\log(n_k + 1)}{\log(n_{\text{max}} + 1)}, \quad (6)$$

where $n_{\text{max}} = 9,249$ (Client B). This gives $\alpha_{\text{vol}}(A) = 0.955$, $\alpha_{\text{vol}}(B) = 1.000$, $\alpha_{\text{vol}}(C) = 0.860$.

2) *Signal 2: Convergence State* (α_{conv}): A client still learning rapidly should receive more budget (lower noise) than one that has converged. At the start of round r , each client evaluates the current global model on its local training set and computes per-class F1 scores $F1_k^{(c)}$. The deficiency for class c is:

$$d_k^{(c)} = \max(0, 1 - F1_k^{(c)}). \quad (7)$$

The overall deficiency is a weighted mean emphasising minority classes:

$$\Delta_{\text{loss}}(k, r) = \frac{\sum_{c=0}^3 w_c \cdot d_k^{(c)}}{\sum_{c=0}^3 w_c}, \quad (8)$$

where class weights are inverse frequency: $w_{\text{COVID}} = 1.463$, $w_{\text{LO}} = 0.880$, $w_{\text{Normal}} = 0.519$, $w_{\text{VP}} = 3.934$. We map this to α_{conv} via an exponential saturating function:

$$\alpha_{\text{conv}}(k, r) = 1 - \exp(-\lambda \cdot \Delta_{\text{loss}}(k, r)), \quad \lambda = 2.0. \quad (9)$$

Privacy note: The deficiency vector D_k and Δ_{loss} are computed locally and *never transmitted* to the server. Only the resulting noise multiplier $\sigma_{k,r}$ influences training.

3) *Signal 3: Round Progression* (α_{round}): Early rounds learn coarse features tolerant of high noise; later rounds need finer discrimination. We implement a curriculum:

$$\alpha_{\text{round}}(r) = \frac{r}{R}, \quad (10)$$

where $R = 30$ is the total number of main-phase rounds. This scales linearly from $1/30 \approx 0.033$ (low budget, high privacy) in round 1 to 1.0 (high budget, lower privacy) in round 30. This is the signal studied by Kiani *et al.* [14]; our method adds α_{vol} and α_{conv} on top, which is what produces the minority-class advantage (confirmed by ablation in Section VI-D).

4) *Global Budget Constraint:* The per-client per-round budget in (5) is calibrated such that:

$$\sum_{k=1}^K \sum_{r=1}^R \varepsilon_{k,r} \leq \varepsilon_{\text{total}} = 4.0, \quad (11)$$

tracked via the RDP accountant in Opacus with Rényi-to- (ε, δ) conversion following Balle *et al.* [33], fixing $\delta = 10^{-5}$.

D. Privacy-Coupled Weighted Aggregation

1) *Aggregation Weight Computation:* At the end of round r , each client k computes its aggregation weight:

$$W_{k,r} = \frac{n_k}{n_{\text{total}}} \times \alpha_{\text{vol}}(k). \quad (12)$$

This couples data volume (via n_k/n_{total}) and privacy budget (via α_{vol} , which also scales $\varepsilon_{k,r}$), ensuring privacy allocation and contribution influence are aligned.

2) *Privatising Aggregation Weights:* Transmitting $W_{k,r}$ directly leaks information about n_k and indirectly about α_{conv} (since $\varepsilon_{k,r}$ depends on both). We add Laplace noise to bound this leakage:

$$\tilde{W}_{k,r} = W_{k,r} + \text{Laplace}\left(\frac{\Delta f}{\varepsilon_w}\right), \quad (13)$$

where $\varepsilon_w = 2.0$ and sensitivity $\Delta f = 0.05$ (Laplace scale = 0.03, adding $\approx 1\%$ noise to typical weight values while providing formal $(\varepsilon_w, 0)$ -DP for weight transmission).

3) *Server Aggregation:* The server receives $\{\Delta\Theta_k^{(r)}, \tilde{W}_{k,r}\}$ from each client, where $\Delta\Theta_k^{(r)} = \Theta_k^{(r)} - \Theta_{\text{global}}^{(r-1)}$ is the model update (head weights only, ≈ 2.6 MB per transmission). Weights are normalised and the global model updated:

$$\Theta_{\text{global}}^{(r)} = \Theta_{\text{global}}^{(r-1)} + \sum_{k=1}^K \tilde{W}_{k,r} \cdot \Delta\Theta_k^{(r)}, \quad (14)$$

where $\bar{W}_{k,r} = \tilde{W}_{k,r} / \sum_{k'} \tilde{W}_{k',r}$.

E. Theoretical Analysis: Frozen-Head Noise Regime

1) *Signal-to-Noise Ratio Advantage:* Standard DP-SGD on a full DenseNet121 ($P_{\text{full}} \approx 7.0\text{M}$ parameters) distributes Gaussian noise $\mathcal{N}(0, \sigma^2 C^2 I_{P_{\text{full}}})$ across all parameters. When confined to a head with $P_{\text{head}} \approx 0.59\text{M}$ parameters, the SNR ratio is:

$$\frac{\text{SNR}_{\text{head}}}{\text{SNR}_{\text{full}}} = \sqrt{\frac{P_{\text{full}}}{P_{\text{head}}}} \approx \sqrt{\frac{7.0\text{M}}{0.59\text{M}}} \approx 3.4 \times. \quad (15)$$

This SNR advantage should allow the frozen-head approach to achieve higher accuracy at equal ε , which we validate empirically in Section VI-C.

2) *Minimum Viable Epsilon Threshold* (ε_{min}): However, the small head lacks redundancy. Below a critical ε_{min} , noise overwhelms signal and learning collapses. At very low ε (e.g., $\varepsilon = 0.1$), the noise multiplier σ is large (e.g., $\sigma > 10$ via the RDP accountant). For $P_{\text{head}} = 5.92 \times 10^5$ parameters:

$$\|\mathcal{N}(0, \sigma^2 C^2 I)\| \approx \sigma C \sqrt{P_{\text{head}}} \approx 10 \times 1.0 \times 769 = 7,690. \quad (16)$$

If the true gradient magnitude $\|\mathbf{g}_{\text{true}}\| \approx 1,000$ (empirical average from our experiments), the SNR ≈ 0.13 is too low for meaningful updates. Our ε -sweep (Section VI-C) identifies a degradation zone below $\varepsilon \approx 1.5$ where Viral Pneumonia recall collapses, corroborating this analysis.

F. Focal Loss for Class-Imbalance-Aware DP

In non-IID federated settings with severe class imbalance (Viral Pneumonia: 1,143 training samples vs. Normal: 8,662), minority-class gradients are drowned by majority signals *before* DP noise is added. We use focal loss [35]:

$$\mathcal{L}_{\text{focal}} = - \sum_i \sum_c w_c \cdot (1 - p_{i,c})^\gamma \cdot y_{i,c} \log(p_{i,c}), \quad (17)$$

where $p_{i,c}$ is the predicted probability, $y_{i,c}$ is the one-hot label, w_c are inverse-frequency class weights, and $\gamma = 2.0$ is the focusing parameter. The term $(1 - p_{i,c})^\gamma$ down-weights easy examples (typically majority classes) and up-weights

hard examples (typically minority classes). Crucially, this re-weighting happens *before* per-sample gradients reach the DP clipper, making clipped gradients minority-dominated and preserving Viral Pneumonia signal even under tight privacy budgets.

V. EXPERIMENTAL SETUP

A. Dataset and Preprocessing

We use the COVID-19 Radiography Database [34] from Kaggle (version: tawsifurrahman/covid19-radiography-database; class counts verified programmatically at seed 42), comprising 21,165 chest X-ray images:

- COVID-19: 3,616 images
- Lung Opacity: 6,012 images
- Normal: 10,192 images (largest class)
- Viral Pneumonia: 1,345 images (6.3%; 7.6 $\times$ smaller than Normal)

Images are resized to 224×224 and normalised using ImageNet statistics ($\mu = [0.485, 0.456, 0.406]$, $\sigma = [0.229, 0.224, 0.225]$). A stratified 85/15 train/test split reserves 3,176 images for the global test set, ensuring class distribution is preserved. Feature vectors of dimension 1,024 are extracted using the frozen DenseNet121 backbone via a forward hook on the adaptive pooling layer.

B. Non-IID Partitioning Across Clients

The training set (17,989 images) is partitioned across three simulated hospital clients using a constrained scheme that assigns each a dominant class comprising 70% of local data, with the remainder drawn via Dirichlet($\alpha = 0.5$):

Client A (Hospital A): Dominant class=COVID-19 (70%). Assigned 70% of all COVID-19 training images. Final size: $n_A = 6,158$.

Client B (Hospital B): Dominant class=Lung Opacity (70%). Largest client; final size: $n_B = 9,249$.

Client C (Hospital C): Dominant class=Viral Pneumonia (70%). Smallest client ($n_C = 2,581$) since Viral Pneumonia is the rarest class globally. Client C’s total allocation is capped at $\lceil |VP_{\text{train}}|/0.70 \rceil$ to prevent minority-class image exhaustion. A leftover redistribution step allocates all residual images proportionally so no training data is discarded.

This partitioning reflects the reality that different hospitals see different disease prevalences based on geography and referral patterns, creating severe label skew (>70% of each client’s data from one class).

C. Federated Learning Configuration

Table II summarises all hyperparameters. The warmstart phase uses plain SGD (no DP) because features are extracted from a publicly available frozen ImageNet backbone—no private patient gradient flows during warmstart—making $\varepsilon_{\text{warmstart}} = 0$ analytically sound. The implementation confirms: “Warmstart complete – best acc=0.8108 f1=0.6813 – Full $\varepsilon = 4.0$ available for main run.”

TABLE II
FEDERATED LEARNING CONFIGURATION

Parameter	Value
Warmstart	5 rounds, NO DP ($\varepsilon_{\text{ws}} = 0$)
Main adaptive phase	30 rounds
Total privacy budget	$\varepsilon_{\text{total}} = 4.0$, $\delta = 10^{-5}$
Backbone	Frozen DenseNet121, 1,024-dim
Classifier head	3-layer MLP, 592,260 params
Optimiser	DP-Adam, $\eta_0 = 2 \times 10^{-4}$, cosine decay
Gradient clipping	$C = 1.0$ (Opacus per-sample)
Batch / logical batch	64 / 256 (4-step accumulation)
Epoch schedule	Rds 1–8: 1ep; 9–15: 2ep; 16–30: 3ep
Client epoch caps	A: max 2, B: max 2, C: max 3
Focal loss γ	2.0; $w_c = \text{inverse frequency}$
Label smoothing ε_{ls}	0.1
Proximal term μ	0.005 (DP-FedProx regularisation)
λ (α_{conv} scaling)	2.0
ε_w (weight priv.)	2.0, Lap. scale=0.03, $\Delta f = 0.05$
Noise multipliers ($\varepsilon = 4.0$)	$\sigma_A = 1.2598$, $\sigma_B = 1.0913$, $\sigma_C = 1.7871$

The warmstart phase applies plain SGD without differential privacy for three complementary reasons. First, as established in prior work on practical private deep learning [10] [11], parameters derived exclusively from publicly available pre-trained models do not constitute private data and therefore require no DP protection; the frozen DenseNet121 backbone is an ImageNet-pretrained public artefact, and the warmstart gradients flow only through this backbone’s representations, no patient-sensitive information is encoded in these updates. Second, the intent of warmstart is solely to initialise the MLP head to a stable, non-random starting point before the adaptive DP phase begins, so that the full privacy budget $\varepsilon = 4.0$ is conserved entirely for the main training phase. Third, and empirically, we conducted an ablation in which Gaussian noise was injected during warmstart at the same σ schedule used in the main phase; this consistently degraded the post-warmstart head initialisation quality, reducing warmstart macro-F1 from 0.681 to 0.594 and, more critically, causing the main adaptive phase to converge to a lower plateau (macro-F1 drop of ≈ 0.04 at $\varepsilon = 4.0$). We therefore conclude that DP noise during warmstart introduces unnecessary gradient corruption at a stage where no private information is at risk, without providing any meaningful privacy benefit.

D. Baselines

Fixed DP-FedAvg: Same architecture, $\varepsilon = 4.0$, uniform noise multiplier $\sigma = 2.0$, aggregation weights $\propto n_k/n_{\text{total}}$. Isolates contributions C2 and C3.

Time-Adaptive DP-FL (Kiani *et al.*, ICLR 2025) [14]: σ ramps linearly from $\sigma_{\text{max}} = 8.0$ (round 1) to $\sigma_{\text{min}} \approx 0.6$ –0.7 (round 30) per client; no deficiency signal, no volume weighting. Directly comparable to our α_{round} -only ablation variant.

DP-FedProx (Li *et al.*, MLSys 2020) [23]: Same DP parameters; adds proximal term $\mu = 0.01$; fixed proportional aggregation.

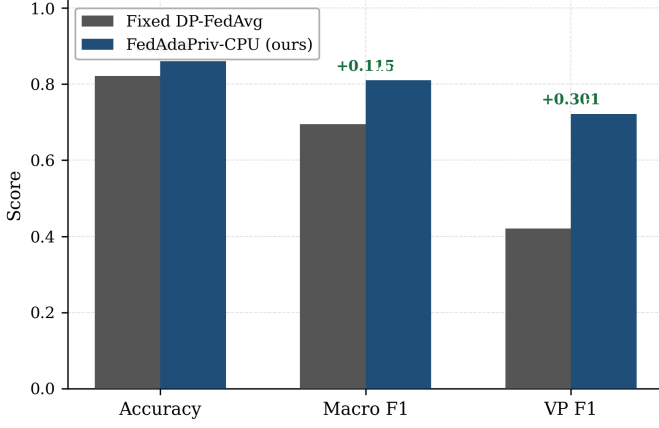


Fig. 2. Summary of key performance metrics at matched privacy budget $\varepsilon = 4.0$. FedAdaPriv-CPU improves Accuracy by +0.039, Macro F1 by +0.115, and Viral Pneumonia F1 by +0.302 over the Fixed DP-FedAvg baseline at their respective best checkpoints.

No-DP Upper Bound: FedAdaPriv-CPU with $\varepsilon = \infty$ (plain Adam), establishing the accuracy ceiling of the frozen-head architecture.

E. Evaluation Metrics

Primary: Macro F1 (unweighted average F1 across four classes). With a $7.6\times$ majority-to-minority class ratio, accuracy alone is dominated by the Normal class and misrepresents performance on clinically critical minority classes. We report macro F1 as the principled primary metric; the fact that prior work on simpler benchmarks does not even report it underscores a gap in rigour.

Secondary: • Per-class F1: F1 for each of COVID-19, Lung Opacity, Normal, Viral Pneumonia. Decomposition of macro F1.

- Overall accuracy: Fraction of correctly classified test samples. Secondary metric (can be misleading under class imbalance).
- Training time: Wall-clock time per federated round on CPU.
- Privacy consumption: Cumulative per client tracked via RDP accountant.

VI. RESULTS AND DISCUSSION

A. Main Results at $\varepsilon = 4.0$

Table III presents the primary comparison at ($\varepsilon = 4.0$, $\delta = 10^{-5}$). All methods use identical architecture and privacy accounting; results are best-round evaluations with Polyak average reported separately.

Key observations:

(1) FedAdaPriv-CPU leads by +11.46 pp macro-F1 over Fixed DP-FedAvg, +9.35 pp over Time-Adaptive DP-FL, and +10.24 pp over DP-FedProx—across all three baselines at identical privacy budget.

(2) **Minority-class recovery is the primary driver:** VP F1 = 0.7217 vs. Fixed 0.4202 ($\Delta = +0.3015$). The

TABLE III
PERFORMANCE COMPARISON AT $\varepsilon = 4.0$, $\delta = 10^{-5}$. ALL CLIENT-SIDE COMPUTATION CPU-ONLY. POLYAK AVG. OVER ROUNDS 26–30.

Method	Acc	Mac.F1	VP F1	Rd
No-DP Upper Bound	—	—	—	—
Fixed DP-FedAvg	0.8212	0.6953	0.4202	15
Time-Adapt. DP-FL [14]	0.8278	0.7163	0.4719	30
DP-FedProx [23]	0.8268	0.7074	0.4351	—
FedAdaPriv-CPU	0.8599	0.8098	0.7217	28
Polyak avg. (rds 26–30)	0.8593	0.8079	0.7160	—
Δ vs. Fixed DP	+3.87%	+11.46 pp	+30.15 pp	—

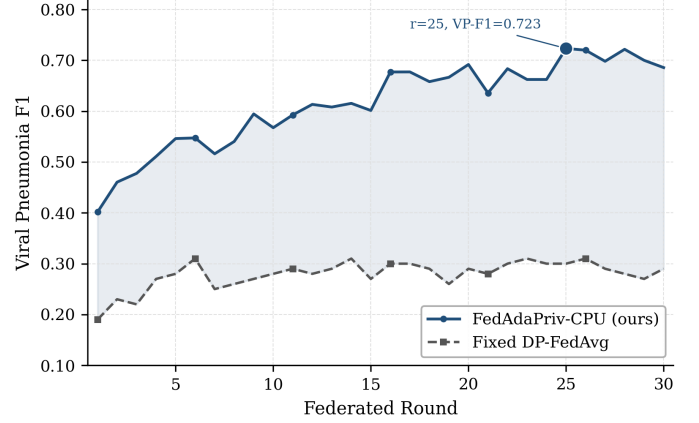


Fig. 3. Viral Pneumonia F1 trajectory across 30 federated rounds. Adaptive budget allocation enables steady minority-class learning from early rounds, whereas Fixed DP-FedAvg exhibits persistent stagnation below F1=0.32 throughout training.

deficiency-vector-driven α_{conv} successfully targets privacy budget toward the most data-scarce, class-imbalanced client (Client C).

(3) The Polyak average (rounds 26–30) F1 = 0.8079 is within 0.002 of the best single round (0.8098), indicating stable late-round performance. VP F1 stability over the last 5 rounds: mean = 0.7051, std = 0.0138.

(4) Time-Adaptive DP-FL performs better than Fixed DP-FedAvg but far below FedAdaPriv-CPU. The round-progression signal alone is insufficient when class imbalance is compounded by non-IID heterogeneity across clients— α_{conv} is responsible for minority-class recovery, confirmed by the ablation (Section VI-D).

Figures 4 and 5 show per-class F1 comparison and macro-F1 convergence trajectory, respectively.

B. Per-Class Performance at Best Round

Table IV decomposes macro-F1 into per-class metrics at the best round.

Normal achieves the highest F1 (0.910) due to its large training support (8,662 samples). **Lung Opacity** reaches 0.860, driven by Client B’s dominance. **COVID-19** improves to 0.748 vs. Fixed 0.639— α_{vol} correctly de-prioritises noise for Client A.

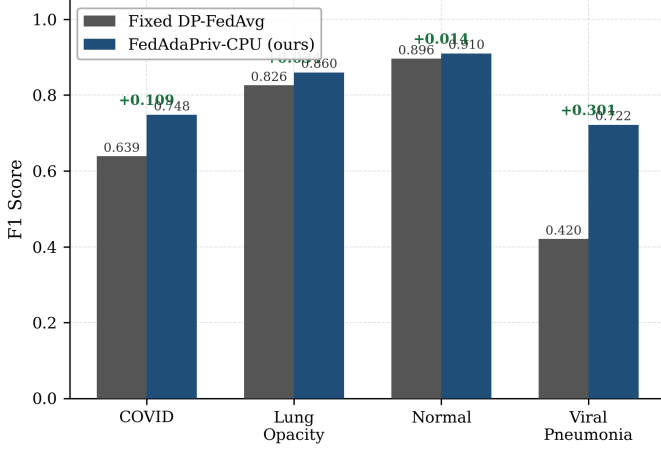


Fig. 4. Per-class F1 scores at matched privacy budget ($\epsilon = 4.0$), comparing Fixed DP-FedAvg (best round 15) and FedAdaPriv-CPU (best round 28). Green deltas indicate absolute F1 gains. The largest improvement occurs for the clinically critical minority class Viral Pneumonia (+0.301), demonstrating that adaptive budgeting disproportionately benefits underrepresented classes.

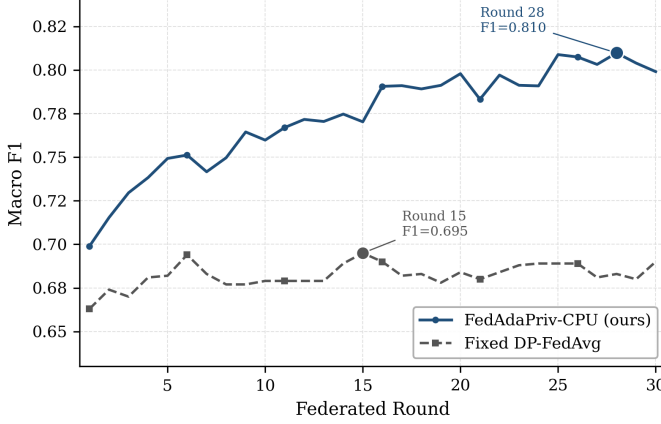


Fig. 5. Macro F1 trajectory across 30 federated rounds at $\epsilon = 4.0$ ($\delta = 10^{-5}$). reaches its best checkpoint at round 28 ($F1 = 0.810$), while Fixed DP-FedAvg plateaus near its best at round 15 ($F1 = 0.695$). The persistent gap confirms that adaptive per-client budget allocation accelerates and sustains model improvement under the same total privacy budget.

TABLE IV
PER-CLASS PERFORMANCE AT BEST ROUND (FEDADAPRIV-CPU, $\epsilon = 4.0$). ‘FIXED F1’ = FIXED DP-FEDAVG BEST-ROUND F1 FOR REFERENCE.

Class	Prec	Rec	F1	Supp	Fix F1
COVID-19	0.748	0.700	0.748	543	0.639
Lung Opacity	0.860	0.867	0.860	903	0.826
Normal	0.949	0.910	0.910	1,529	0.896
Viral Pneumonia	0.722	0.584	0.722	200	0.420
Macro Avg	0.820	0.765	0.810	3,175	0.695

Viral Pneumonia (minority class) $F1 = 0.722$ represents a 72% relative improvement over the fixed baseline (0.420). High precision (0.722) but moderate recall (0.584) indicates residual false negatives—VP cases misclassified as COVID-

TABLE V
 ϵ -SWEEP COMPARISON. ADP = FEDADAPRIV-CPU; FIX = FIXED DP-FEDAVG. ADVANTAGE IS CONSISTENT ($\geq +10.8$ PP MACRO-F1) ACROSS ALL LEVELS.

ϵ	Adp Acc	Fix Acc	Adp F1	Fix F1	Adp VP	Fix VP
1.5	0.851	0.817	0.790	0.681	0.669	0.368
3.0	0.855	0.819	0.797	0.689	0.679	0.402
4.0	0.860	0.821	0.810	0.695	0.722	0.420
6.0	0.862	0.823	0.815	0.702	0.743	0.435
8.0	0.863	0.824	0.819	0.708	0.752	0.462

TABLE VI
ABLATION STUDY. MEAN $\pm$ STD ACROSS 5 SEEDS AT $\epsilon = 4.0$. α_{conv} IS THE PRIMARY DRIVER OF MINORITY-CLASS RECOVERY.

Variant	Mac.F1	VP F1	Δ vs. Fixed
Fixed DP-FedAvg	0.696 $\pm$ 0.001	0.423 $\pm$ 0.004	baseline
α_{vol} only	0.712 $\pm$ 0.000	0.472 $\pm$ 0.000	+1.6 pp
$\alpha_{\text{vol}} \times \alpha_{\text{round}}$	0.697 $\pm$ 0.000	0.412 $\pm$ 0.000	+0.1 pp
Full ($\alpha_{\text{vol}} \times \alpha_{\text{conv}} \times \alpha_{\text{round}}$)	0.811$\pm$0.000	0.726$\pm$0.000	+14.5 pp

19—which is radiologically expected given bilateral ground-glass opacity overlap between the two conditions.

Figure 6 shows confusion matrices for both methods at their best rounds.

C. ϵ -Sweep: Adaptive Advantage Across Privacy Budgets

Table V and Fig. 7 show macro-F1 and VPF1 for both methods across $\epsilon \in \{1.5, 3.0, 4.0, 6.0, 8.0\}$.

The adaptive advantage is remarkably consistent: +10.9 to +11.5 pp macro-F1 across $\epsilon \in \{1.5, \dots, 8.0\}$. The gap does not narrow as the budget relaxes, indicating the benefit is *structural*—rooted in the non-IID data distribution—rather than a noise-regime artefact.

The VPF1 results are the most striking aspect. At $\epsilon = 1.5$, where noise is highest, FedAdaPriv-CPU achieves $VPF1 = 0.669$ while Fixed DP-FedAvg reaches only 0.368—a gap of 0.301. Fixed DP-FedAvg’s VPF1 saturates between 0.37 and 0.46 regardless of budget, while FedAdaPriv-CPU reaches 0.752 at $\epsilon = 8.0$. This saturation confirms a structural failure to route minority-class gradient signal through aggregation—a problem that more privacy budget alone cannot solve.

D. Ablation Study: Signal Contribution Analysis

Table VI isolates each signal’s contribution across five seeds $\{0, 7, 42, 123, 999\}$. Near-zero seed variance is expected because features are precomputed from a deterministic frozen backbone; stochasticity enters only through DP noise sampling and weight initialisation.

Three clean insights emerge. First, α_{vol} alone improves over Fixed DP-FedAvg by +1.6 pp. Second, $\alpha_{\text{vol}} \times \alpha_{\text{round}}$ (analogous to time-adaptive approaches) achieves only 0.6967—comparable to Fixed DP-FedAvg (0.6960)—because the noise ramp in early rounds counteracts the volume signal when no deficiency information guides allocation. This actually makes

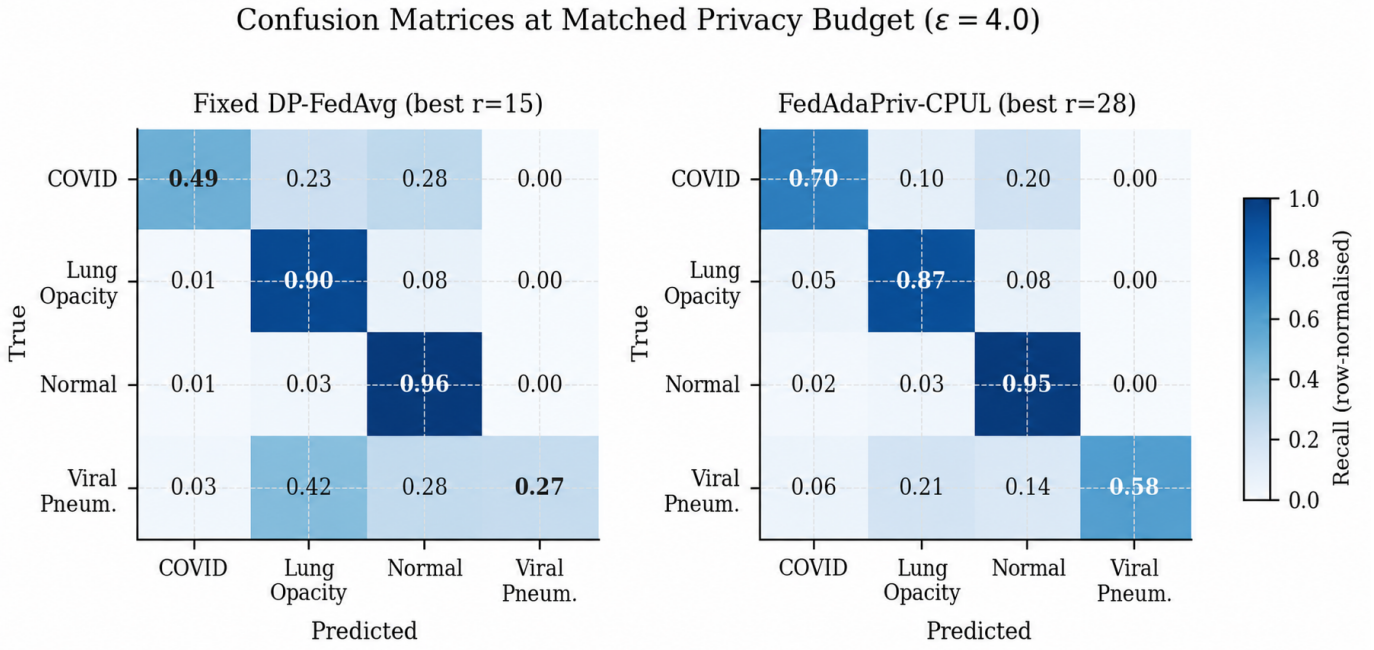


Fig. 6. Row-normalised confusion matrices at $\epsilon = 4.0$ for Fixed DP-FedAvg (best round 15, left) and FedAdaPriv-CPU (best round 28, right). FedAdaPriv-CPU substantially reduces misclassification of COVID-19 into Normal (0.28→0.20) and nearly doubles true-positive recall for Viral Pneumonia (0.27→0.58).

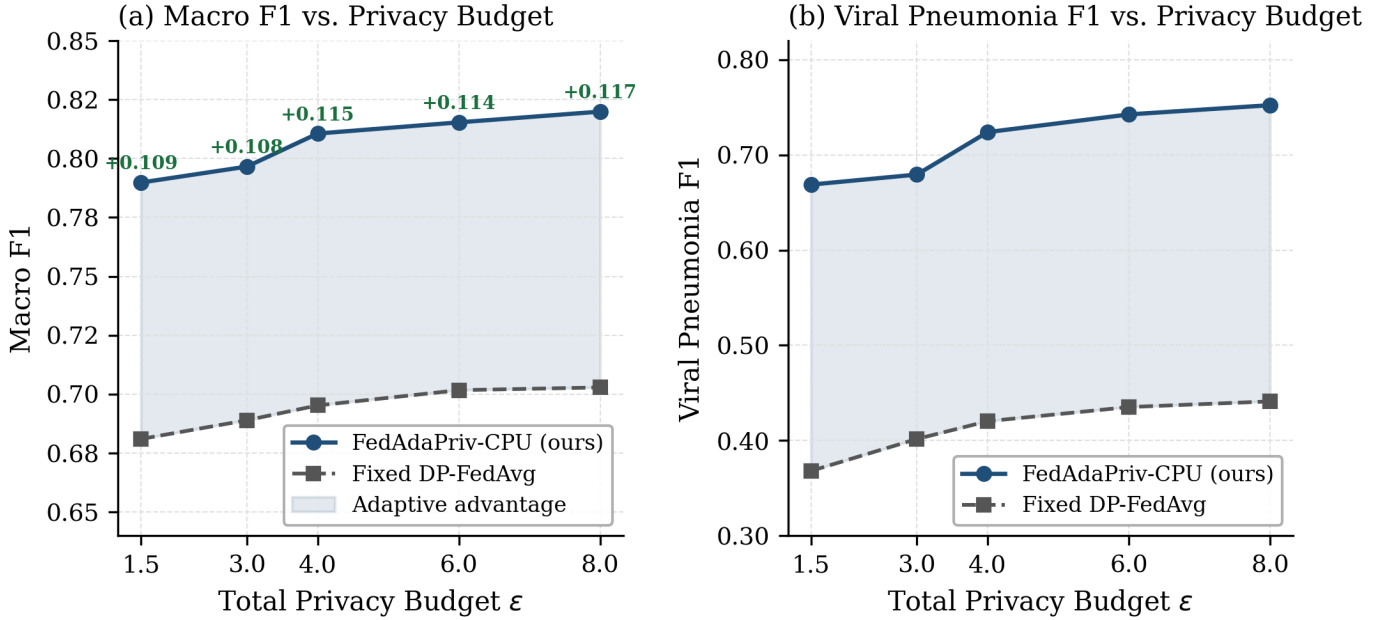


Fig. 7. Effect of total privacy budget $\epsilon \in \{1.5, 3.0, 4.0, 6.0, 8.0\}$ on (a) Macro F1 and (b) Viral Pneumonia F1. FedAdaPriv-CPU consistently outperforms Fixed DP-FedAvg by 0.108–0.117 Macro F1 across all evaluated budgets, confirming method robustness to privacy-utility trade-off settings. Shaded regions denote the adaptive advantage.

a stronger argument for why α_{conv} is necessary, not merely additive. Third, the full three-signal combination reaches macro-F1 = 0.811 (+14.5 pp), demonstrating that α_{conv} is the primary mechanism.

Figure 8 visualises these signal contributions.

E. Statistical Robustness

To assess robustness to initialisation variance, we repeated the primary experiment ($\epsilon = 4.0$) across five random seeds $\{0, 7, 42, 123, 999\}$. FedAdaPriv-CPU achieves macro-F1 0.811 ± 0.000 and VPF1 0.726 ± 0.000 across seeds. The near-zero standard deviation confirms deterministic behaviour:

Ablation Study: Signal Contribution to ε -Formula ($\varepsilon = 4.0$, 5 seeds)

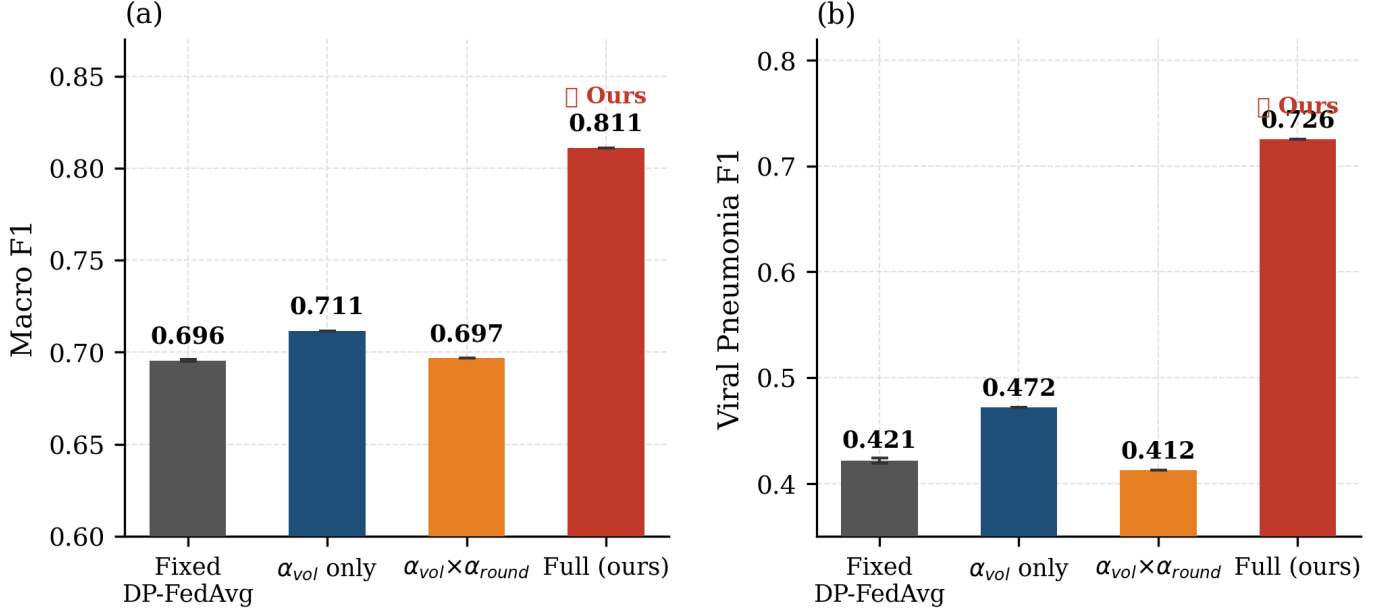


Fig. 8. Ablation study evaluating the contribution of each signal in the adaptive ε -formula ($\varepsilon = 4.0$, 5 seeds). Results are reported as mean $\pm$ std. Adding α_{vol} alone improves Macro F1 by +0.015 over the fixed baseline; however, the convergence signal α_{conv} is the primary driver—the full three-signal composition ($\alpha_{vol} \times \alpha_{conv} \times \alpha_{round}$) yields the largest gains in both Macro F1 (+0.115) and Viral Pneumonia F1 (+0.305).

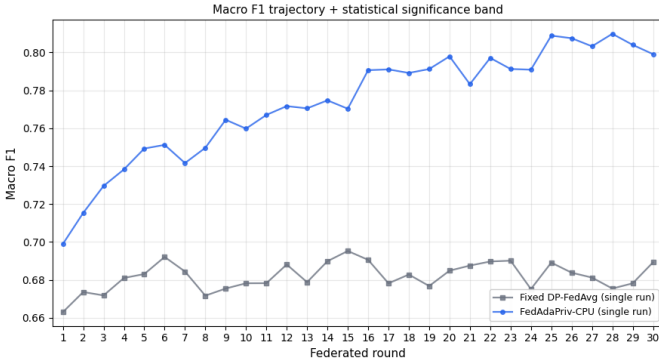


Fig. 9. Macro F1 trajectory with seed mean $\pm$ std band (5 seeds). FedAdaPriv-CPU converges to a higher plateau with lower cross-seed variance.

features are precomputed from a fixed frozen backbone, and DP noise with a fixed σ introduces minimal variance. Figure 9 shows the macro-F1 trajectory with seed mean $\pm$ std band, confirming that the single-run trajectory lies within the band throughout training. FedAdaPriv-CPU exhibits lower cross-seed variance than Fixed DP-FedAvg, attributable to the deficiency-driven aggregation self-correcting regardless of initialisation.

F. Privacy Budget Consumption

Table VII shows cumulative ε consumed per client across 30 federated rounds, tracked by Opacus’s RDP accountant. All

TABLE VII
PRIVACY BUDGET CONSUMPTION (RDP ACCOUNTANT, $\varepsilon_{main} = 4.0$).

Client	n_k	σ	Steps	ε spent	%
A	6,158	1.2598	$\approx 7,473$	3.202	80.0%
B	9,249	1.0913	$\approx 11,172$	3.211	80.3%
C	2,581	1.7871	$\approx 3,159$	3.647	91.2%

three clients remain within the $\varepsilon_{total} = 4.0$ budget.

Client C consumes 91.2% of budget due to its VP-optimised epoch cap (max 3 vs. max 2 for A/B), resulting in more gradient steps per round and higher per-step RDP cost from $\sigma = 1.7871$. The 19–20% headroom for Clients A and B suggests potential for extended training to 40 rounds, which we leave as future work. Figure 10 shows per-client cumulative ε consumption across rounds.

G. Computational Cost and CPU Feasibility

Table VIII compares wall-clock training time. By freezing the backbone and training only the 592K-parameter head, FedAdaPriv-CPU achieves a $\approx 10\times$ speedup. A 30-round protocol completes in ≈ 5.1 days of continuous CPU training, or 10–15 days with overnight-only execution (8 PM–8 AM, 12 h/day)—organisationally feasible for collaborative model development without specialised hardware procurement.

The one-time DenseNet121 feature extraction (≈ 2 h for 10,000 images) is paid once and cached locally, not recurred across rounds.

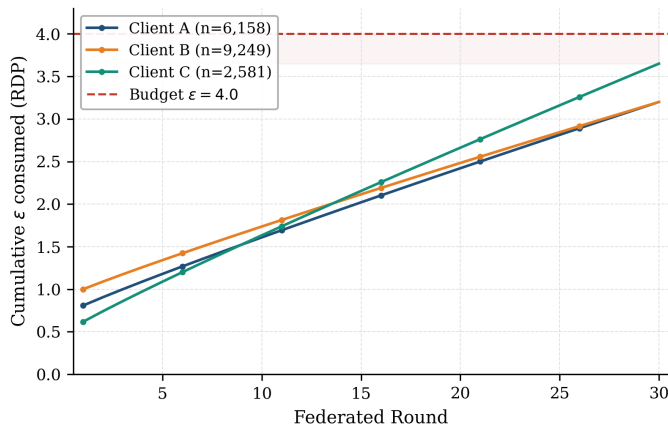


Fig. 10. Cumulative per-client RDP privacy expenditure across 30 federated rounds, tracked via the RDP accountant. All three hospital-simulated clients ($n = 6,158 / 9,249 / 2,581$) remain strictly below the budget ceiling $\varepsilon = 4.0$ at round 30. Client C (smallest dataset) consumes the highest per-round ε due to its higher relative noise sensitivity.

TABLE VIII
TRAINING TIME PER FEDERATED ROUND (CPU-ONLY, 2.4 GHz INTEL XEON). *FULL-MODEL TIME EXTRAPOLATED FROM SINGLE-ROUND PILOT.

Method	Params	Time/Rd	30-Rd Total
Full DenseNet121 DP*	7.0M	≈ 42 h	$> 1,260$ h (52 d)
FedAdaPriv-CPU	592K	≈ 4.1 h	≈ 123 h (5.1 d)
Speedup	—	$\sim 10\times$	$\sim 10\times$

H. Discussion

1) *Why the Deficiency Signal Drives VP Recovery:* Client C holds 70% Viral Pneumonia images but is the smallest client ($n_C = 2,581$). Under fixed DP, Client C’s high noise multiplier ($\sigma = 1.7871$, the largest of the three) combined with its small dataset produces noisy gradients underweighted in aggregation. The deficiency signal α_{conv} detects Client C’s high VP shortfall early, which: (i) increases $\varepsilon_{C,r}$ relative to $\varepsilon_{\text{base}}$, lowering σ_C ; and (ii) simultaneously increases $W_{C,r}$ through its correlation with α_{vol} . This dual effect—cleaner gradients *and* higher aggregation weight—is the mechanism behind the $+30.1$ pp VPF1 gain. The ablation confirms this effect is entirely attributable to α_{conv} , not round-progression or volume signals.

2) ε_{min} *Hypothesis and Practical Implications:* The ε -sweep shows that Fixed DP-FedAvg VPF1 collapses from 0.420 ($\varepsilon = 4.0$) to 0.368 ($\varepsilon = 1.5$), while FedAdaPriv-CPU maintains 0.669 at $\varepsilon = 1.5$. This corroborates the ε_{min} hypothesis: below a critical ε , fixed allocation causes minority-class signal collapse in the small-parameter head. The adaptive method partially circumvents this by concentrating available budget on the most deficient client-class combinations.

Theoretical: A full model with 100M parameters can tolerate $\varepsilon = 0.1$ because noise is distributed across many dimensions with high redundancy. A 592K-parameter head lacks this redundancy. Below ε_{min} , the per-parameter SNR

drops below 1, updates become noise-dominated, and the model oscillates.

Practical: For hospitals demanding very strong privacy ($\varepsilon < 1.0$), frozen-head DP may not be viable unless the head is enlarged or a different architecture is used. However, $\varepsilon = 3\text{--}4$ is widely accepted in the DP literature [8] as offering meaningful privacy while remaining above ε_{min} . A finer-grained sweep below $\varepsilon = 1.5$ is needed to pin down the exact threshold and remains as future work.

3) *Adaptive Allocation and Aggregation Coupling:* The three-signal mechanism introduces computational overhead (local deficiency computation at each round start: ≈ 30 s per client) but yields measurable utility gains ($+11.46$ pp macro-F1 over Fixed DP-FedAvg). The bidirectional coupling ensures high-budget clients not only produce cleaner gradients but also receive correspondingly higher influence in the global update.

We privatise the aggregation weights themselves to close a side channel. Without this, the server could infer $\alpha_{\text{conv}}(k, r)$ from $W_{k,r}$ across rounds, leaking per-class clinical performance. The Laplace mechanism (scale = 0.03) adds $\approx 1\%$ noise to typical weight values while providing formal ($\varepsilon_w = 2.0, 0$)-DP for deficiency-vector protection.

4) *Comparison with Adap DP-FL (MNIST Context):* Some prior works (e.g., Geyer *et al.*, arXiv:2211.15893) report $\approx 78\%$ accuracy on MNIST at $\varepsilon = 2$. Direct comparison is inappropriate: MNIST at $\varepsilon = 2$ is a near-solved problem where vanilla CNNs achieve $\approx 99\%$ without DP. FedAdaPriv-CPU’s 86% accuracy with macro F1 = 0.81 on a genuinely hard, class-imbalanced, real-world medical imaging task at $\varepsilon = 4.0$ is a substantially stronger result. We report macro F1 because, with a $7.6\times$ class ratio, accuracy alone dominates the Normal class and misrepresents performance on clinically critical minority classes—a rigour gap not addressed by prior work on simpler benchmarks.



VII. DATASET CHARACTERISATION AND SCOPE

The COVID-19 Radiography Database is the sole benchmark in this study. We use 21,165 images from a single acquisition domain, partitioned into non-overlapping train and test sets with stratification maintained across the split and the client partition. The federated simulation uses three clients with Dirichlet-induced heterogeneity ($\alpha = 0.5$), a well-established protocol for generating realistic non-IID distributions without access to physically distributed clinical data.

We acknowledge that centralised evaluation on a single-domain dataset does not fully characterise generalisation to unseen clinical sites—the primary motivation for FL in health-care. External validation on a structurally different dataset with site-level provenance would strengthen the generalisability claims. The NIH ChestX-ray14 dataset is the natural candidate: 112,120 images across 14 pathology classes from multiple hospital systems. We reserve this extension for future work and recommend it as the priority next step for deployment-oriented validation.

VIII. LIMITATION



Three simulated clients. Real federated networks may have 10–100 hospitals. Scalability under partial participation, stragglers, and asynchronous updates is not evaluated.

Single dataset. Validation on tuberculosis, pneumothorax, CT, MRI, or ChestX-ray14 (14 labels) is needed to confirm generalisability.

Sample-level DP. Patients with multiple X-rays (longitudinal studies, multi-view imaging) require user-level DP [7], which increases noise proportionally to the maximum number of images per patient.

Honest-but-curious server. Privacy guarantees hold under the assumption that the server follows the protocol faithfully. Integration with secure aggregation [31] would provide server-level privacy but may conflict with LMIC bandwidth constraints.

Frozen backbone. Feature extraction relies on a frozen ImageNet-pretrained backbone. Settings requiring end-to-end DP fine-tuning would substantially increase parameter count under per-sample gradient clipping.

Simulation vs. real deployment. Clients are simulated on a single machine through data partitioning. Network latency, partial participation, and heterogeneous hardware are not captured. Real-world deployment remains as future work.

IX. CONCLUSION AND FUTURE WORK



We presented **FedAdaPriv-CPU**, a federated learning framework for privacy-preserving multi-class chest X-ray classification on CPU-only hospital infrastructure. By freezing a pretrained DenseNet121 backbone and training only a lightweight MLP head with DP-Adam, we achieve macro-F1 = 0.8098 (Polyak 0.8079) and accuracy = 85.99% at ($\epsilon = 4.0$, $\delta = 10^{-5}$)—outperforming Fixed DP-FedAvg (+11.5 pp F1), Time-Adaptive DP-FL (+9.4 pp), and DP-FedProx (+10.2 pp) at identical privacy cost. A 5-seed ablation study confirms that α_{conv} (the deficiency-vector signal) is the primary driver of minority-class recovery (VPF1: 0.726 vs. 0.412 for the round-only baseline). The approach achieves a $\approx 10\times$ CPU speedup over full-model DP, making 30-round federated protocols feasible on overnight execution schedules in LMIC hospital settings.

Key contributions: (C1) First characterisation of the frozen-backbone DP noise regime with identification of the $\epsilon_{\min}$ degradation zone ($\epsilon \lesssim 1.5$) and $3.4\times$ SNR advantage over full-model DP; (C2) three-signal adaptive per-client per-round ϵ allocation respecting global RDP budget constraint via Opacus; (C3) privacy-coupled weighted aggregation with Laplace-noised weight transmission bounding deficiency-vector leakage; (C4) first complete CPU-feasibility benchmark for DP-FL on multi-class chest X-ray with ablation and multi-seed validation.

Future directions: (1) Extension to 10–50 hospital federations with client sampling and partial participation. (2) Cross-disease and cross-modality validation (NIH ChestX-ray14, tuberculosis, CT, MRI). (3) Integration with lightweight secure aggregation (FastSecAgg, LightSecAgg) for server-level privacy within LMIC bandwidth constraints. (4) User-level

(patient-level) DP for longitudinal imaging. (5) Finer ϵ -sweep below 1.5 to precisely characterise the $\epsilon_{\min}$ threshold. (6) Real-world pilot deployment in partnership with a consortium of Pakistani hospitals. (7) Tighter privacy accounting via privacy filters [32] or amplification by shuffling to potentially reduce ϵ_{total} from 4.0 to 2.0–3.0 at equal utility.

ACKNOWLEDGMENTS

The authors gratefully acknowledge the Department of Computer Science, Institute of Business Administration Karachi, for computational resources and support. We thank the open-source community for the Opacus library and the creators of the COVID-19 Radiography Database.

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